

MUHAMMAD AHMAD DUGGAL

AI/ML Engineer | Backend Developer | Python & MERN Stack Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science student with hands-on experience in Machine Learning, Backend Development, and data-driven systems. Skilled in building end-to-end ML pipelines using Python, scikit-learn, XGBoost, and TensorFlow, as well as scalable backend applications using Node.js, Express.js, MySQL, and MongoDB. Strong foundation in Data Structures, Algorithms, Database Design, OOP, and Software Engineering. Seeking an AI/ML or Backend Engineering internship to contribute technical and problem-solving skills in a professional environment.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C++, SQL

AI / ML: scikit-learn, XGBoost, TensorFlow, PyTorch, Feature Engineering, Classification, Regression, Model Evaluation, Time-Series Analysis, RAG

Backend & APIs: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, Stripe Integration

Databases: MySQL, MongoDB, ERD Modeling, Schema Normalization (1NF–3NF)

Tools & Core CS: Git, GitHub, Postman, VS Code, Jupyter Notebook, Linux, DSA, OOP, System Design

PROJECTS

Trading Intelligence System | *Python · XGBoost · scikit-learn · RAG · Feature Engineering*

- Built a multi-component AI trading system: Smart Money Concepts (SMC) pattern detector, an XGBoost classifier for trade signal probability, and a RAG-based knowledge engine for context-aware decision support.
- Engineered 50+ features from raw OHLCV data — technical indicators (RSI, MACD, BB), market structure signals, and order-flow imbalance — pipeline designed for live Binance Futures inference.

MERN Stack Restaurant Marketplace | *React.js · Redux Toolkit · Node.js · MongoDB · Stripe · JWT*

- Built a fully responsive multi-role food ordering platform (Customer / Restaurant / Admin) with component-based architecture and REST API-driven backend.
- Implemented Redux Toolkit for global state management, dynamic cart, real-time order updates, JWT auth, RBAC, and Stripe payment simulation.

AutoSOS — Crash Detection System | *MySQL · REST APIs · JWT · Information Security*

- Designed a normalized relational database (1NF–3NF) for real-time accident reporting with JWT auth and information security best practices.

Smart Inventory Management System | *C++ · OOP · DSA*

- Built a modular CLI system using OOP principles (Inheritance, Polymorphism), Linked Lists, and file-based persistence with robust exception handling.

EDUCATION

Bachelor of Science — Computer Science, Government College University (GCU), Lahore

2022 – 2026

Relevant Coursework: Data Structures & Algorithms · Design & Analysis of Algorithms · OOP · Database Systems · Operating Systems · Machine Learning · Probability & Statistics

CERTIFICATIONS

- Salesforce Administrator Training — Trailhead